

EVALUATION RESULTS

System Performance and Solution Effectiveness

Federated and explainable crop-yield analytics · GRU over 12-month climate sequences · federation by state

10,223

rows after cleaning

501

districts

20

states as clients

1990–2015

years

Every figure in this deck is read from `artifacts/results/*.json` by `scripts/09_evaluation_deck.py` at build time. No number is typed into a slide by hand, so the deck, the report and the results API cannot disagree.

What is being compared

Arm	Kind	What it does
Global mean	Baseline	One mean yield for the whole panel
District mean	Baseline	Each district's own historical mean
District trend	Reference	Linear extrapolation of each district's trend
Centralised	Ceiling	One model, all data pooled in one place
Local-only	Floor	One model per state, no sharing at all
FedPer	Federated	Shares the GRU base, keeps the head local
FedAvg	Federated	Weighted averaging of all weights each round
FedProx	Federated	FedAvg plus a proximal term against client drift

Skill is measured against the district trend baseline, not against zero. A model that cannot beat 'extrapolate each district's own trend' has not earned its complexity. Learned arms are run over 5 seeds (42–46) and reported as mean $\pm$ sample standard deviation.

Model cost

Quantity	Value
Total parameters	16,033
Base (GRU) parameters	13,632
Head parameters	2,401
Shared under FedPer	85%
State dict on the wire	62.6 KB
GRU FLOPs per sample	294,912
Forward cost	$O(T \cdot H^2) = O(12 \cdot 64^2)$
Inference	0.01 ms/sample (batch 200)

WHY THIS MATTERS

At 16,033 parameters the model is small enough that a full state dict is 62.6 KB. That is what makes the federation cheap: the per-round cost is set by model size, not by data size.

85% of the parameters sit in the GRU base. FedPer shares exactly that and keeps the 2,401-parameter head local, which is where its communication saving comes from.

Inference at 0.01 ms/sample means a whole state can be scored in well under a second on CPU.

Communication cost per round

Algorithm	Rounds	KB / round	Total MB	s / round
FedAvg	29	1,189.9	33.7	2.07
FedProx	26	1,189.9	30.2	2.06
FedPer	57	1,011.8	56.3	1.90

15.0% less per round

FedPer against FedAvg, because the head never leaves the client.

THE HONEST READING

Per round FedPer is cheaper. In total it is not: it ran 57 rounds against FedAvg's 29, so it moved 56.3 MB against 33.7 MB overall. The saving is real per-round and disappears once convergence is counted. A deck that quoted only the 15% would be misleading.

FedProx proximal term

μ	Algorithm	R^2	Skill vs trend	L2 vs FedAvg	Rounds
0	FedAvg	0.3019	-0.0918	0.000	26
0.001	FedProx	0.3015	-0.0925	0.509	26
0.01	FedProx	0.2983	-0.0975	1.679	26
0.1	FedProx	0.2940	-0.1042	2.317	28
1	FedProx	0.2865	-0.1160	2.596	32

The proximal term is active, not inert: parameter distance from the FedAvg solution rises monotonically with μ , from 0.509 to 2.596. It is doing what it is supposed to do. It simply does not help here: the best R^2 in the sweep is 0.3019 at $\mu = 0$, and no setting lifts FedProx above the trend baseline. Reporting the sweep matters more than reporting a tuned winner, because it shows the negative result was measured rather than assumed.

Predictive skill against a ceiling and a floor

Arm	R^2	Skill vs trend	Worst-client RMSE	Seeds
Global mean	-0.263	—	2,015	1
District mean	0.304	—	1,189	1
District trend	0.348	baseline	1,148	1
Centralised	0.448 ± 0.018	$+0.136 \pm 0.029$	1,005	5
Local-only	0.391 ± 0.020	$+0.047 \pm 0.031$	1,150	5
FedPer	0.364 ± 0.016	$+0.005 \pm 0.024$	1,301	5
FedAvg	0.295 ± 0.010	-0.103 ± 0.015	1,110	5
FedProx	0.292 ± 0.009	-0.108 ± 0.015	1,113	5

The sentence that must accompany this table. Federated methods trade accuracy for cross-client uniformity. Centralised (0.448) and local-only (0.391) beat FedAvg (0.295) and FedProx (0.292) because state-level heterogeneity dominates this panel; FedPer recovers accuracy (0.364) precisely by declining to average the head. Read without this sentence the table looks like a failed federation. Read with it, the ordering is the expected consequence of strongly non-IID data, and is itself the finding.

Do federated explanations agree with centralised ones?

Comparison	Kendall τ	95% interval	Top-10 Jaccard	Pairs
Seed control (centralised vs itself)	0.719 ± 0.059	[0.684, 0.753]	0.631	10
FedAvg	0.387 ± 0.082	[0.315, 0.460]	0.460	5
FedProx	0.380 ± 0.077	[0.312, 0.448]	0.441	5
FedPer	0.508 ± 0.042	[0.471, 0.545]	0.546	5

The control is the contribution. Two centralised models that differ only by random seed agree at $\tau = 0.719$, not 1.0. That is the ceiling any federated comparison could reach, so a federated τ is only meaningful as a fraction of it. Without this row, every federated number below would look like a failure to reproduce the explanation; with it, the question becomes how much of the achievable agreement survives federation.

AGRONOMIC PLAUSIBILITY • PASS

3 of the top 10 features are monsoon months. Top features: irrigation ratio, nitrogen consumption, gross cropped area, phosphate consumption, gross irrigated area, jul precipitation. Federated attributions use per-client baselines because no client may access another's data; centralised uses the same strategy for a fair comparison.

Do the explanations actually drive the prediction?

Arm	AOPC top-k	AOPC random	Ratio	Verdict
Centralised	0.484	0.215	2.2×	PASS
FedAvg	0.249	0.037	6.8×	PASS
FedProx	0.247	0.036	6.9×	PASS
FedPer	0.367	0.084	4.4×	PASS

WHAT THE TEST DOES

Delete the top-k attributed features and measure how far the prediction moves; delete k features at random and measure the same. If the attribution is real, the first number is much larger. Evaluated on Uttar Pradesh, k up to 15, 284 rows.

THE CAVEAT THAT BELONGS NEXT TO THE TABLE

FedProx has the highest ratio (6.9×) but the lowest random-deletion AOPC (0.036). The ratio is flattering because that model is relatively unresponsive to feature deletion overall, not because its ranking is sharper. Ratios are not comparable across arms with different baseline responsiveness, and the deck does not rank the arms by them.

Why the federation is posed by state

Variable • month	Pooled r	Within-district r	Sign flip
precipitation • nov	+0.306	-0.020	yes
precipitation • oct	+0.278	-0.010	yes
windspeed • dec	+0.116	-0.159	yes
evapotranspiration • dec	+0.283	+0.017	no
evapotranspiration • nov	+0.248	-0.009	yes
precipitation • dec	+0.247	-0.002	yes

32 of 60

month-variable pairs invert sign between the pooled and the within-district correlation.

Pooled across districts, post-monsoon rain looks like the dominant predictor and monsoon rain looks harmful. Both are artefacts: wet districts are disproportionately rainfed and low-yielding, so 'wet' proxies for 'low-yielding' and reverses the apparent sign. Removing each district's own mean restores the agronomic direction.

This is the measured form of the project's claim that heterogeneity carries signal and should be preserved rather than averaged away. It is the empirical justification for partitioning by state rather than pooling.

Against published work

Work	Setting	Reported
This project • centralised	501 districts, strict chronological split	$R^2 = 0.448$
This project • FedPer	Same split, per-state clients	$R^2 = 0.364$
Yazdi et al. [8]	Rainfed cereals, strict chronological validation	$R^2 = 0.271$
De Clercq & Mahdi [6]	247 Indian rice districts, centralised	$R^2 \leq 0.82$
Malashin et al. [14]	Indian state data, optimised deep network	$R^2 = 0.92$
Shams et al. [13]	XAI-CROP, centralised	$R^2 = 0.9415$
Yenkikar et al. [11]	246k records, 33 states, centralised ensemble	$R^2 = 0.9827$
Dey et al. [26]	Federated, 4,513 Maharashtra samples	acc > 99% (not R^2)
Mukherjee & Buyya [27]	Federated, ring/mesh topologies	acc > 93% (not R^2)

Read this honestly. Against loosely-validated centralised studies this project is behind on raw R^2 . Against the one study using comparable strict chronological validation on rainfed data it is ahead. The federated accuracy figures are classification-style accuracy on smaller, more homogeneous samples and are not R^2 , so they are not comparable and are not compared here. The claim this project actually makes is different: no reviewed federated work evaluates against a centralised ceiling, a trend baseline and an explanation-fidelity test at once.

What is not here

Claim	Status
Climate-stress R^2 drop per arm	ood.json not generated
Perturbation response (+2 °C, -20% rain)	perturbation.json not generated
Wilcoxon p-values vs centralised	significance.json not generated
Feature-group ablation	per-group runs not committed

Stages F and G do not exist. An earlier project summary described climate-stress, perturbation and significance results as complete. Neither the evaluation code nor the artifacts are present in the repository or in the local working copy. Those figures have no artifact behind them and are quoted nowhere in this deck.

The feature-group comparison, that neither climate nor agronomic covariates beat the baseline alone and only their combination does, is the strongest single argument available in R3. The script writes each run to its own file; the runs simply need to be executed and committed before the claim can be quoted.

Artifacts behind this deck: ablation, agreement, attributions, baselines, complexity, confound, federation_history, fidelity, heterogeneity, meta, mu_sweep.